

PROJECT DOCUMENTATION

Functional Specification Document

Employee Data Management using ServiceNow Import Sets and Transform Maps

1. Introduction

This Functional Specification Document (FSD) describes the functional behaviour of the completed ServiceNow employee-data management project. It is adapted from the supplied project documentation format, while replacing the MERN-specific sections with the components actually used in this project. The document is intended to give a reader a clear understanding of what the system does, how the data flows, how the functionality is configured, and how the completed solution can be tested.

1.1 Project Title

Employee Data Management using ServiceNow Import Sets and Transform Maps

1.2 Scope

The project covers importing employee information from an Excel spreadsheet into ServiceNow, staging the incoming data in an Import Set, transforming the source fields into the Employee Test table, matching existing employees using Employee ID through Coalesce, and presenting the resulting employee information through reports and an Employee Analytics Dashboard.

2. Project Overview

The project provides a structured alternative to repeatedly entering or manually reconciling employee information. The administrator loads the spreadsheet into ServiceNow, runs the configured transformation, and reviews the target records. When an Employee ID already exists, the Coalesce configuration is used to identify the existing employee so changed information can be updated rather than creating a second employee record.

2.1 Purpose

The purpose is to make employee-data loading and maintenance more consistent by separating data intake, staging, field mapping, record matching and reporting.

2.2 Key Features

Excel employee-data import through ServiceNow Load Data / Import Set.

Employee Import staging table for incoming records.

Transform Map for source-to-target field mapping.

Employee ID used as the Coalesce matching value.

Insertion of new employee records.

Update of existing employee records when matching data changes.

Employees by Department report.

Employees by Location report.

Employee List Report.

Employee Analytics Dashboard.

3. Architecture

The functional architecture follows a staged import pattern. The spreadsheet is the source, the Import Set table acts as the staging layer, the Transform Map performs the mapping into the Employee Test table, and Coalesce provides the matching rule used during transformation. Reports and the dashboard provide the presentation layer.

3.1 Functional Data Flow

Administrator prepares the employee spreadsheet.

Spreadsheet is loaded into ServiceNow through Load Data.

Incoming rows are stored in the Employee Import staging table.

The Transform Map maps source fields to Employee Test.

Coalesce checks Employee ID against existing employee records.

A new Employee ID results in an insert; a matching Employee ID allows the existing record to be updated with changed source values.

Reports and the dashboard present the resulting employee data.

4. Setup Instructions

Unlike the supplied MERN format, this project does not require Node.js, MongoDB, npm commands or application-server environment variables. The required environment is a ServiceNow instance with access to the relevant table, import and reporting configuration.

4.1 Configuration Sequence

Create or verify the Employee Test table and required fields.

Create or verify the Employee Import staging table.

Load the employee spreadsheet through Load Data.

Create/configure the Transform Map and field mappings.

Set Employee ID as the Coalesce field.

Run the transformation and inspect the target records.

Create/open the employee reports and Employee Analytics Dashboard.

5. Folder / Project Structure

Because this is a ServiceNow configuration project rather than a source-code application, the repository structure is organized around documentation and evidence instead of client/server source folders.

6. Running / Using the Application

There is no local frontend/backend start command for this project. The solution is used inside the configured ServiceNow instance.

Open the ServiceNow instance and navigate to the configured data-import area.

Use Load Data to upload the employee spreadsheet.

Review the Import Set / Employee Import staging records.

Run the configured Transform Map.

Verify inserted and updated records in Employee Test.

Open the configured reports and Employee Analytics Dashboard.

7. Functional Interface / Configuration Documentation

The project does not expose a custom REST API as part of the documented scope. The primary functional interfaces are the ServiceNow Load Data/import interface, Transform Map configuration, Employee Test table, reports and dashboard.

8. Authentication and Authorization

The project uses the access controls of the ServiceNow environment. The supplied project documentation does not specify custom authentication, token handling or a separate authorization service. Therefore, no custom token/session mechanism is claimed in this FSD. Access to import configuration, employee records, reports and dashboard should be limited according to the roles and permissions configured in the ServiceNow instance.

9. User Interface

The main user-facing areas are ServiceNow's Load Data/import interface, the transformation configuration/results, the Employee Test record view, the configured reports, and the Employee Analytics Dashboard.

Add the actual ServiceNow screenshots to the GitHub repository under the project evidence folder; screenshots are not fabricated in this document.

10. Testing

Testing is organized around functional correctness, data integrity, user acceptance and relevant performance checks. The project documentation already defines UAT scenarios for new employee insertion, existing employee update, repeated unchanged import, field mapping, reports and dashboard. Performance checks focus on import processing, transformation, matching and report/dashboard usability.

11. Screenshots / Demo Evidence

For GitHub documentation, include screenshots that demonstrate the actual implementation rather than generic ServiceNow images.

Load Data / Import Set upload and result.

Employee Import staging records.

Transform Map field mappings.

Transform execution/result.

Employee Test table with inserted employee records.

Existing employee record before and after an update using Employee ID matching.

Employees by Department report.

Employees by Location report.

Employee List Report.

Employee Analytics Dashboard.

12. Known Issues

The supplied project documentation does not provide a formal defect register. The following are therefore documented as scope limitations rather than invented defects:

No production-scale performance benchmark is established.

No external enterprise integration is documented.

Actual UAT sign-off must be completed by the project/faculty reviewer.

Repository screenshots and measured performance results should be added when available.

Access control details beyond the ServiceNow environment are not specified in the supplied project material.

13. Future Enhancements

Add additional employee fields and validation rules.

Introduce scheduled or recurring imports where appropriate.

Add stronger validation and error-handling/reporting for rejected source rows.

Extend dashboard metrics and reporting.

Introduce role-based access aligned with organizational responsibilities.

Evaluate larger data volumes and record actual import/transform performance.

Add additional data sources or integrations if future requirements call for them.

Document Control

End of Functional Specification Document

Field	Details
Team ID	SWTID-2026-8740
Project Title	Employee Data Management using ServiceNow Import Sets and Transform Maps
Team Members	Project Team – individual member names/roles not supplied in the source material
Platform	ServiceNow
Document Version	1.0
Date	03 October 2026

Component	Functional Responsibility
Excel Employee Data	Source of Employee ID, Employee Name, Email, Department and Location.
Load Data	Accepts the spreadsheet and starts the ServiceNow import process.
Employee Import	Stores incoming source rows before transformation.
Transform Map	Maps source fields to the Employee Test target fields.
Coalesce / Employee ID	Determines whether an incoming employee corresponds to an existing target record.
Employee Test	Stores the transformed employee records.
Reports	Present employee information by department, location and list view.
Employee Analytics Dashboard	Combines configured employee reports into one review view.

Requirement	Project Requirement
ServiceNow instance	A ServiceNow environment where the project configuration can be created and tested.
User access	Permission to load data, manage the import configuration, run transformations and view reports/dashboard.
Source file	Employee spreadsheet containing the fields required by the Transform Map.
Target table	Employee Test table with the required employee fields.
Import configuration	Employee Import staging table and configured Transform Map.
Matching rule	Employee ID configured as the Coalesce field.

Repository Area	Contents
Project Design Phase	Problem–Solution Fit, Proposed Solution and Solution Architecture documents.

Repository Area	Contents
Project Planning Phase	Planning Logic and Project Planning documents.
Project Development Phase / Performance Testing	Performance testing plan and evidence.
Project Development Phase / User Acceptance Testing	UAT Functional Specification and UAT Execution & Report.
Evidence / Screenshots	ServiceNow screenshots showing import, transformation, records, reports and dashboard.
Source Data / Sample Files	Project-approved sample spreadsheet, if permitted for repository sharing.

Interface / Component	Input	Processing	Output
Load Data / Import Set	Excel employee spreadsheet	Loads source rows into staging.	Employee Import records.
Transform Map	Staged employee rows	Maps source fields to target fields and applies matching.	Transformed Employee Test records.
Coalesce	Employee ID	Matches incoming Employee ID with existing target data.	Insert or update behaviour.
Department Report	Employee Test data	Groups/summarizes employee data by department.	Department report.
Location Report	Employee Test data	Groups/summarizes employee data by location.	Location report.
Employee List Report	Employee Test data	Displays employee records.	Employee list view/report.
Analytics Dashboard	Configured reports	Combines configured report widgets.	Employee Analytics Dashboard.

UI Area	User Action	Expected Outcome
Load Data	Select and upload the employee spreadsheet.	Import process is created and source rows are staged.
Transform Map	Run the transformation after staging data.	Employee records are inserted or updated according to the mapping and Coalesce rule.
Employee Test	Review transformed records.	Employee information is available in the target table.
Reports	Open department, location or list report.	Employee information is presented in the selected report.
Employee Analytics Dashboard	Open dashboard.	Configured employee reports are visible in one place.

Test Area	Coverage
Import	Spreadsheet can be loaded into the Import Set.
Transformation	Transform Map executes and maps required employee fields.
Data Matching	Employee ID correctly identifies an existing employee.
Insert	A new Employee ID creates a new target record.
Update	Changed values for an existing Employee ID update the existing record.
Repeated Import	Unchanged repeated data does not create unnecessary duplicates.
Reporting	Department, location and employee-list reports display the resulting data.
Dashboard	Employee Analytics Dashboard displays the configured reports.

Version	Date	Description	Prepared For
1.0	03 October 2026	Initial Functional Specification Document for the completed project.	Team SWTID-2026-8740